

DNA Process Station (PCS)

The **Valmet DNA Process Station (PCS)** is a central component in the Valmet DNA Distributed Control System (DCS), responsible for executing control logic and interfacing with field devices.

Roles and features:

- Modified Linux OS
- Connects to the Valmet DNA process network using redundant Ethernet.
- Communicates using a Valmet-specific protocol over UDP/IP
- Can be used in systems ranging from tens to thousands of I/O points.

1. Real-Time Process Control = CPU

- Executes control applications such as:
 - Basic process control
 - Advanced control algorithms (e.g., fuzzy logic, MPC)
 - Executes **Function Block CAD (FbCAD)** designed control logic
 - Executes **Sequence CAD (SeqCAD)** designed sequences (DCS systems 2)

2. Field Device Communication, FBC = Field Bus Controller

- Interfaces with various field devices via standard fieldbus protocols:
 - ACN IO (MIO (Metso IO)), Centralised IO (Rack IO)
 - PROFINET
 - Profibus
 - OPC / OPC UA
 - Modbus (RTU / TCP/IP)
- Manages I/O signals and ensures accurate data exchange between sensors, actuators, and the control system

Different hardware versions:

ACN = Application and Control Node



ACN MR (Modular Rail), Deprecated version ACN SR1

- Small form factor
- Limited scalability
- DP66 lab GP01



ACN CS, Deprecated version ACN C20

- Industrial PC
- Better scalability
- DP39 lab EP01 and EP02



ACN RT

- Industrial PC
- Better scalability
- Most powerful option
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Backup Station (BU)

Roles and features:

- Not a backup station in the literal sense
- Windows OS
- Connects to the Valmet DNA process network using redundant Ethernet.
- Communicates using a Valmet-specific protocol over UDP/IP

1. System Software Distribution

- During software download action from EAS, **all nodes** load their system software through the BU server.
- This ensures consistent and centralized deployment of system components.

2. Application Hosting

- The BU server contains **all applications** required by different nodes
- After booting, all nodes retrieve their applications and configuration from the BU server.

3. Automatic Recovery

- The system can **automatically restart** after a power failure without manual intervention, thanks to the BU server's role in reinitializing nodes

4. Centralized Management

- Acts as a **central node** for managing software and application versions across the network.

Different hardware versions:



ACN AS (Application Server), ACN PO (Process Operation), ACN CSW

Or can be virtualized as done in JAMK system

Alarm Station (ALS, ALP)

The **Valmet DNA Alarm Station (ALS)** is a specialized component within the Valmet DNA automation system, designed to manage and distribute alarms across the control environment.

Roles and features:

- Windows OS

1. Centralized Alarm Management

- Collects and manages alarms from various system components (controllers, field devices, network elements)
- Distributes Alarms to Operating stations (OPS)
- Ensures that alarms are **timestamped** and **categorized** for clarity and traceability

2. Alarm Hierarchy and Prioritization

- Organizes alarms into a **hierarchical structure**, allowing operators to quickly identify critical issues.
- Supports **state synchronization**, ensuring that alarm statuses are consistent across the system

Different hardware versions:



ACN AS (Application Server), ACN PO (Process Operation), ACN CSW

Or can be virtualized as done in JAMK system

Engineering Activity Server (EAS)

The **Valmet DNA Engineering Activity Server (EAS)** is a engineering component of the Valmet DNA automation system, designed to support engineering, configuration, and maintenance activities across the plant lifecycle. EAS is the main user interface in this course.

1. Centralized Engineering Database

- Hosts all control application (in FunctionBlock CAD format), system configurations, and documentation.
- Enables **multi-user access** for concurrent engineering and maintenance tasks

2. Lifecycle Management

- Supports full lifecycle management of:
 - Control applications
 - Field buses (e.g., PROFINET, Profibus, Modbus)
- Through **DNA Field Device Manager**, EAS enables:
 - Efficient commissioning
 - Proactive maintenance planning
 - Real-time device diagnostics
- Ensures consistent documentation and version control

3. Graphical Design Tools

- Integrates with tools like:
 - **Function Block CAD (FbCAD)** for control logic design
 - **Sequence CAD** for sequence programming
 - **Graphics Designer** for HMI and process visualization
- All designs are stored and managed within the EAS repository / database

4. Function Testing

- Provides tools to test control applications before deployment.
- Live data visualization in CAD diagrams helps validate logic and performance

5. Network and Cabinet Design

- Includes **DNA Network Designer** for:
 - System layout
 - Cabinet layouts
 - I/O labeling and connection diagrams
- Supports WYSIWYG drag-and-drop design and direct device access from diagrams

Common features for all Nodes / Stations:

Slots

In the **Valmet DNA Distributed Control System (DCS)**, the term **"Slot"** refers to **logical or physical positions within the system architecture** where specific automation components or functions are configured or executed.

Back in the days of Damatic XD and VME bus, stations consisted of physical cards, each with its own function (CPU, NCU, FBC, BU, etc.)



Now days slots still exist, but in virtualised form:

C:\dna\64a1.sys - Slot Controller - DNA Program Starter

Standard slots		Extended slots			
Slot	Status	Autostart	Program	Username	Title
01	RUNNING	YES	C:\dna\CA\BU\buc_16.11.exe		GB01
02	RUNNING	YES	C:\dna\QA\use\program\cdmgr.exe		G109
03	RUNNING	YES(LOGON)	C:\dna\QA\use\client\qemgr.exe	dna	QEMGR
04					
05					
06	RUNNING	YES	C:\dna\QA\alp\program\als.exe		G1A1
07					
08	RUNNING	YES	C:\dna\EA\LoopAutoTuner\lib\rmixcom.exe		Communication Server for Loop...
09					
10					
11					
12					
13					
14					
15	RUNNING	YES	C:\dna\Shared\ncu2\ncu2sb_9.23.exe		NCU2-15
16	RUNNING	YES	C:\dna\Shared\ncu2\ncu2sb_9.23.exe		NCU2

☐ All slots

Start Stop Show Slot Log Show Starter Log Reread Sysfile Exit

GB01, G1O9, G1A1

BU, Operate, ALS

64a1 37

192.168.45.59

WIN-RTS-TOKEN 3

G1O9



D202733UA : Virtualization Valmet DNA Remote User Server (Amount: 1)

D202734UA : Virtualization Valmet DNA Remote User (Amount: 35)

D200004UA : Back-up base licence, per node

D201767UA : DNA Operate Alarms & Events

1x1Core

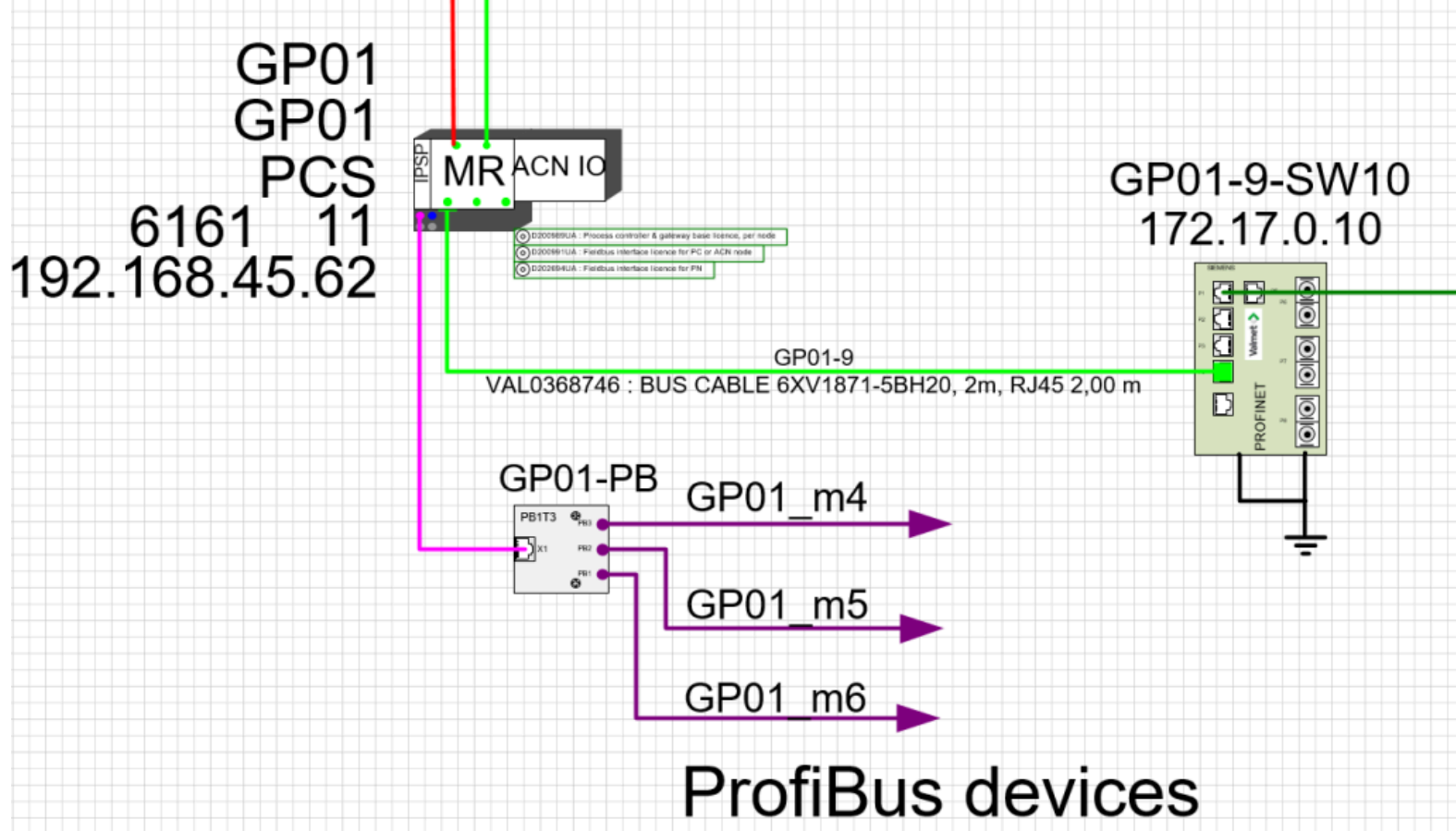
8GB

50GB

6161,GP01 6161,GP01 616

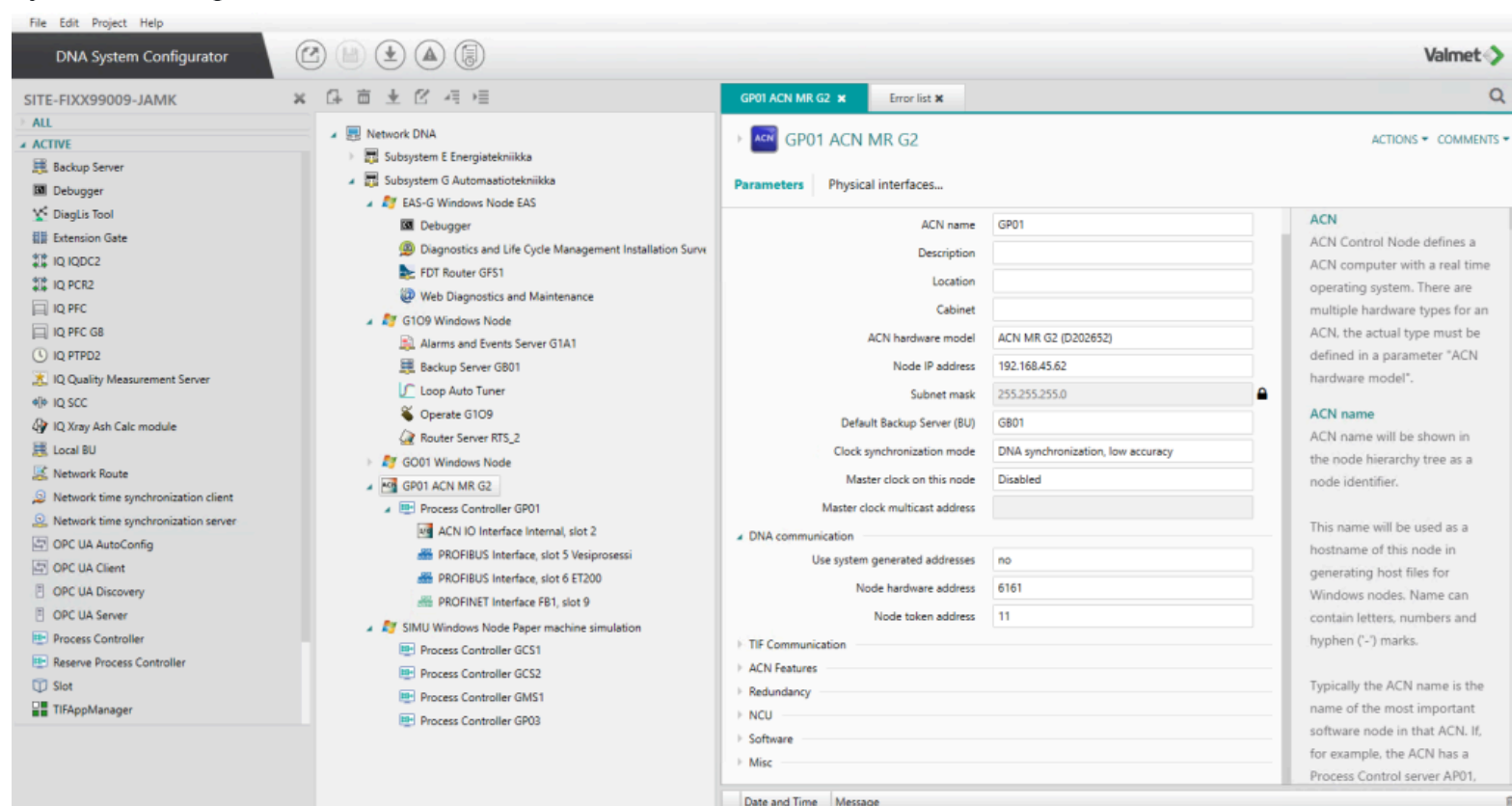
Slot	Status	Autostart	Program	Title
01	RUNNING	YES	pcs_8.116_386	GP01
02	RUNNING	YES	fbcl_9.25_386	GP01_m2
03				
04				
05	RUNNING	YES	fbcl_9.25_386	GP01_m5
06	RUNNING	YES	fbcl_9.25_386	GP01_m6
07	RUNNING	YES	ibc_9.90_386	IBC 0
08				
09	RUNNING	YES	fbcl_9.25_386	GP01_m9
10				
11				
12				
13	STOPPED	NO	dpmon_2.26_386	dpmon
14	STOPPED	NO	pnmon_1.1_386	pnmon
15				
16	RUNNING	YES	ncu2s_9.23_386	NCU2

Start Stop ☐ All Slots Extended slots Exit



As you can see, Nodes can be combined and scaled quite freely --> Flexibility

System Configurator



Valmet DNA Station / Node Naming Principle

1. Subsystem Identification

Each station name begins with a **subsystem code**, which helps identify the part of the plant or process the station belongs to. For example:

- A – Subsystem A (e.g., a specific production line or area)
- B – Subsystem B

- **C** – Subsystem C
These letters are typically assigned based on plant layout or functional grouping.
- In JAMK, we are using
 - **G - Subsystem** for Automation lab DP66 activities, and
 - **E - Subsystem** for Energy technology lab DP39
- Subsystems can consist of several **control rooms**, eg **G1, G2, E1, E2** etc

2. Station Type Code

The second character in the name indicates the **type of station**:

- **P** – Process Controller Station (PCS)
- **A** – Alarm Station (ALS)
- **B** – Backup Station (BU)
- **E** – Engineering Activity Server (EAS)
- **O** – Operator Station (OPS)
- **D** – Diagnostic Station (DIA)
- **O** – OPC Station (OPS)

3. Station Number

The numeric part (e.g., **01, 02, 10**) identifies the **specific instance** of the station within the subsystem. This helps distinguish between multiple stations of the same type.

4. Example Breakdown

Let's take the name **GP01**:

- **G** – Subsystem G
- **P** – Process Controller
- **01** – Station number 01

Other examples:

- **GB01** – Subsystem G, Backup station, number 01
- **G1A1** – Subsystem G, Control room G1, Alarm station, number 1
- **G109** – Subsystem G, Control room G1, Operator station, number 9
- **EB01** – Subsystem E, Backup station, number 01
- **EAS-G** – Subsystem G, Engineering Activity Server (EAS)

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◀ DCS In general

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HW Engineering ▶

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